

Perceptually Optimized Chromatic Power Saving in XR Using Binocular Asymmetry

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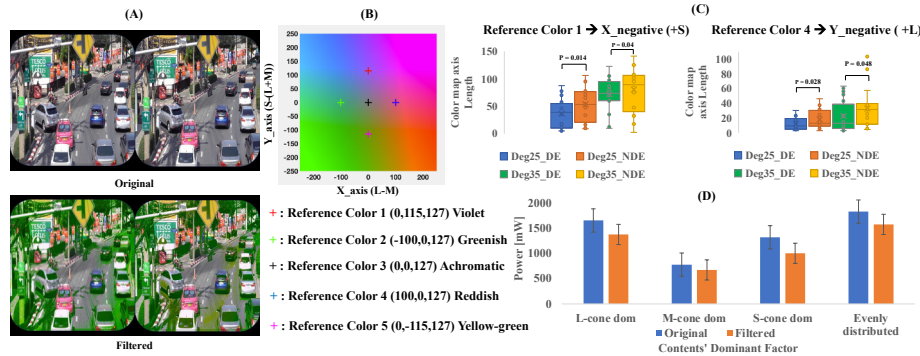


Figure 1: (A) Original and binocular-asymmetric chromatic reduction examples. (B) Reference color directions and per-eye discrimination maps in DKL space. (C) Dominant- (DE) and non-dominant-eye (NDE) color tolerance. (D) Resulting power savings across video contents grouped by dominant chromatic components (L-, M-, S-cone dominant, evenly distributed).

ABSTRACT

High-resolution Extended Reality (XR) rendering imposes energy demands. While perceptually driven power optimization methods exploit properties of the human visual system, they assume symmetric visual sensitivity across both eyes. We investigate whether inter-ocular asymmetry in chromatic sensitivity can be leveraged for power reduction. Using 4-Alternative Forced Choice (4AFC) measurements and Radial Basis Function Neural Network modeling (RBFNN), we derived per-eye discrimination maps. These maps were discretized and validated on video content using 2-Alternative Forced Choice evaluation. Results show that the non-dominant eye tolerates stronger chromatic reduction without degrading binocular quality, enabling up to 24% power savings.

Index Terms: Binocular-asymmetric rendering, Chromatic perception, Perceptual power optimization, Extended reality displays.

1 INTRODUCTION

Display power consumption is a major bottleneck for Extended Reality (XR) headsets, where high-resolution binocular rendering demands substantial energy. Perceptually driven optimization techniques, such as foveated rendering, reduce this cost by exploiting properties of the human visual system [1]. However, most existing approaches assume symmetric visual sensitivity across both eyes, overlooking inter-ocular asymmetry and binocular suppression.

Although prior work has examined chromatic perception and quality assessment in immersive displays [2, 3], binocular vision is often treated as perceptually uniform, despite evidence that ocular dominance influences visual processing, particularly under high-quality viewing conditions, and extends to chromatic perception as well [4, 5]. Motivated by this gap, we investigate whether asymmetric chromatic reduction between the dominant eye (DE) and non-dominant eye (NDE) can be perceptually tolerated and exploited for more effective power optimization in XR rendering.

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2 METHODS

2.1 4-Alternative Forced Choice

2.1.1 Stimuli and Apparatus

The experiment was implemented in Unity 2022.3.8f1 and displayed on a Varjo XR-4 headset (native resolution 3840×3744 per eye, 90 Hz). The display was photometrically calibrated for luminance and chromatic accuracy, and sessions were conducted in a dim environment ($\leq 10\text{lx}$) to reduce ambient light interference. Head and fixation stability were monitored using the XR-4's integrated eye-tracking system. A total of 20 participants (9 male, 11 female; mean age = 22.5 years, SD = 2.42) completed the 4-Alternative Forced Choice (4AFC) experiment. Fourteen participants were right-eye-dominant (RED) and six were left-eye-dominant (LED), as determined by the Miles test.

2.1.2 Stimuli and Color Space

Chromatic stimuli were defined in the Derrington Krauskopf Lennie (DKL) color space using a D65 white point and Smith-Pokorny cone fundamentals. Following prior study that samples chromatic sensitivity along the cardinal DKL axes [2], five reference colors were selected from the isoluminant plane ($L = 127$), as illustrated in Fig. 1-(B).

Stimuli appeared as circular patches (3° visual angle) on a gray background and were presented for 500 ms with a fixation dot.

2.1.3 Procedure

Ocular dominance was determined using the Miles test, and DE and NDE were tested in counterbalanced blocks. Participants performed a 4AFC chromatic detection task across three eccentricities (10° , 25° , 35°) [2]. In each trial, four locations were arranged at the eccentricity, with one containing a target differing from the reference color. A 1-up/2-down adaptive staircase method was employed for DKL direction. The initial offset was suprathreshold (± 50 – 60 units), with step sizes decreasing from 8 to 2 units after the second reversal. Each staircase terminated after 10 reversals, and thresholds were computed as the mean of the final six. Trials with fixation deviations exceeding 1.5° were repeated.

2.2 Color Map Generation and Validation

2.2.1 Color Map Generation via RBFNN

Perceptual thresholds from the 4AFC task were modeled using an Radial Basis Function Neural Network (RBFNN), which estimates minimum, mean, and maximum color map per eye [2]. To enable finer-grained perceptual control in video-based validation, intermediate levels were interpolated between these estimates, resulting in six discrete reduction levels along each DKL axis.

2.2.2 Display Power Modeling & Optimization

Power consumption was profiled on a binocular OLED module (Wisecoco, 1080 × 1200 per eye) using an INA219 current sensor, as OLED power consumption varies with pixel-wise chromatic intensity. Fifty-two diagnostic colors were measured, and total power was modeled as a linear function of gamma-corrected RGB intensities: $P_{total} = P_R R_{lin} + P_G G_{lin} + P_B B_{lin} + P_{circuit}$.

Coefficients were estimated via least-squares regression for each panel [6]. For an original color c_{orig} with DKL coordinate t_{dkl} , the RBFNN defines an eccentricity-dependent discrimination ellipse $\mathcal{E}(x_{dkl}) = 0$. The power-optimal perceptually equivalent color is obtained by: $\min_{x_{dkl}} P_{total}(M_{dkl2rgb}(x_{dkl}))$ s.t. $\mathcal{E}(x_{dkl}) \leq 0$.

The closed-form Lagrange multiplier solution was implemented as a real-time compute shader for gaze-contingent rendering [2][6].

2.2.3 Video-Based Psychophysical Validation

To validate how 4AFC-derived color reduction maps affect perceptual quality in realistic content, we conducted video-based validation using a temporal 2AFC paradigm. Four video scenarios were selected to isolate the effect of chromatic reduction while controlling spatiotemporal complexity, satisfying ITU-T P.910 criteria for Spatial Information (30–50), Temporal Information (10–20), and mean Lightness (40–50) [7]. Videos were categorized by dominant DKL chromatic direction (L-, M-, or S-cone dominant, or evenly distributed when no single component dominated) and presented as 12-second segments (3840 × 2160, 60 fps).

In the 2AFC task, pairwise judgments were converted into Just-Objectable-Difference (JOD) using Bayesian pairwise comparison scaling [3], providing a continuous measure to determine the maximum perceptually acceptable reduction level ($JOD > -1$).

To establish a binocular baseline for subsequent asymmetric evaluation, a symmetric reduction condition was first tested with one group of 10 participants (mean age = 24.1, SD = 2.37), who directly compared original videos against symmetrically reduced videos generated from the DE map. A separate group of 12 participants (mean age = 24.2, SD = 1.75; RED = 8, LED = 4) then evaluated asymmetric reduction by comparing videos rendered with the symmetric baseline against videos with additional reduction applied to one eye, using the same JOD criterion. In this phase, reduction was applied to the fellow eye using its corresponding per-eye map, testing both DE-reduced and NDE-reduced configurations.

3 RESULTS

To examine inter-ocular differences in chromatic discrimination, thresholds from the 4AFC task were compared between the DE and NDE using paired t-tests for each DKL direction and eccentricity. As shown in Fig. 1-(C), thresholds were significantly higher in the NDE along the S-cone direction at 25° eccentricity ($p = 0.014$, $Cohen's d \approx 0.53$), 35° eccentricity ($p = 0.04$, $Cohen's d \approx 0.38$) and along the L-cone direction at 25° eccentricity ($p = 0.028$, $Cohen's d \approx 0.45$), 35° eccentricity ($p = 0.048$, $Cohen's d \approx 0.4$). No significant differences were observed at 10° eccentricity or for achromatic stimuli. Consistent with these results, RBFNN-derived discriminability ellipses were broader for the NDE in S- and L-cone regions, supporting the need for per-eye color-map modeling. All

reported statistics are based on 95% confidence intervals. For multiple comparisons across DKL directions and eccentricities, p-values were adjusted using Holm–Bonferroni correction.

Perceptual evaluation using 2AFC with JOD scaling first established a symmetric reduction baseline, defined as the highest of six discrete reduction levels derived from the DE map that maintained non-objectionable quality ($JOD > -1$). This symmetric threshold was reached at Level 1–3, depending on chromatic dominance (L: 3, M: 1, S: 2, evenly distributed: 2).

Using this baseline as a reference, asymmetric reduction revealed a clear inter-ocular difference. When additional attenuation was applied to the DE, perceptual quality consistently dropped below the objectionability criterion ($JOD \leq -1$) at exactly the symmetric reduction threshold (L: 3, M: 1, S: 2, evenly distributed: 2)—indicating that DE-targeted asymmetric reduction provided no additional tolerance beyond symmetric reduction. Whereas applying the same or stronger attenuation to the NDE remained perceptually acceptable at one higher level (L: 4, M: 2, S: 3, evenly distributed: 3). Consequently, asymmetric reduction targeting the NDE achieved higher power savings than symmetric reduction and asymmetric reduction targeting the DE across all chromatic conditions, as estimated using the proposed display power model: L-cone (16.78% vs. 9.72%), M-cone (13.51% vs. 9.61%), S-cone dominant (24.0% vs. 14.1%), and evenly distributed content (13.83% vs. 10.09%).

4 CONCLUSION

We demonstrated that chromatic sensitivity exhibits systematic inter-ocular asymmetry, producing distinct perceptual color maps for the DE and NDE. This asymmetry enables substantially stronger chromatic reduction in the NDE without compromising binocular visual quality. By leveraging this property, our asymmetric rendering strategy achieved superior power savings—up to 24% in S-cone dominant scenarios—surpassing the limits of symmetric approaches. These findings validate binocular-asymmetric, perceptually informed optimization as a critical pathway for next-generation, energy-efficient XR displays.

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